Denys Shkliarenko

ML Engineer | MLOps | Agentic Systems

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PROFESSIONAL SUMMARY

ML Engineer with 5+ years of experience delivering production-grade ML systems, LLM-powered services, and scalable MLOps infrastructure. Proven track record building end-to-end pipelines - from data ingestion and model training to deployment and monitoring - across on-prem, GCP and AWS. Hands-on expertise in agentic AI systems, RAG pipelines, multi-modal LLMs, and API-driven microservices. Experienced in CI/CD automation, containerization, and cloud-native deployment. Effective collaborator across engineering, data science, and product teams.

TECHNICAL SKILLS

Languages: Python, SQL

ML & Data: PyTorch, Scikit-learn, Pandas, NumPy, Dask, PySpark

LLM & GenAI: Anthropic API, OpenAI API, Google Gemini, LiteLLM, HuggingFace, RAG, Agentic Pipelines

MLOps: DVC, Airflow, Kubeflow, Vertex AI Pipelines, GitHub Actions, Docker, Kubernetes, Helm, Terraform

Cloud: GCP (Vertex AI, BigQuery, GCS, Artifact Registry), AWS (S3, EC2, ECR)

APIs & Serving: FastAPI, Asyncio, Streamlit, Gradio

Monitoring: ELK, Prometheus, Grafana, GCP Monitoring

Testing: PyTest, Unittest

PROFESSIONAL EXPERIENCE

ML Engineer - Ecentria Inc. (OpticsPlanet.com)

Feb 2021 - present

ML Engineering & MLOps

- Contributed to a recommendation system (LightFM, BERT4Rec, FAISS, FastAPI) serving real-time personalized suggestions; participated in feature engineering, experimentation, CI/CD setup, refactoring, and progressively took ownership - confirmed improvement in CTR via A/B tests.
- Played an MLOps role on a demand forecasting platform (LightGBM, Dask, Airflow, DVC); implemented CI/CD, optimized Docker images, and led migration of distributed workloads from GCP to AWS including contribution to the dask-cloudprovider open-source library.
- Fine-tuned and deployed transformer-based LLMs using HuggingFace and Vertex AI with DVC-based reproducible versioning.
- Built and maintained CI/CD workflows for ML systems using GitHub Actions, Docker, and Kubernetes; automated infrastructure provisioning with Terraform across GCP and AWS.
- Implemented monitoring and logging solutions for deployed models and services (Elasticsearch, Kibana, GCP Monitoring, Prometheus, Grafana).

LLM & Agentic Systems

- Built and maintained a FastAPI microservice for ID document validation using multimodal LLMs (via LiteLLM), covering OCR field extraction, fuzzy name matching, and PII-safe document handling.
- Implemented production pipelines for LLM-based product specification enrichment using LLM with web grounding, async fan-out and checkpoint.
- Contributed to an agentic AI pipeline (Google ADK) that autonomously researches product categories and generates citation-backed HTML articles at scale, combining Vertex AI Search and live web retrieval.
- Contributed to a document processing service (AWS Lambda, Step Functions, Anthropic API, Amazon Textract) handling OCR, classification, extraction, and validation; refactored the polling-based OCR flow to an event-driven SQS callback pattern, reducing latency and cost.
- Created agentic AI tooling layer for Claude Code for the data analytics team, standardizing how analysts run ad-hoc analyses and build reports end-to-end

Software & API Development

- Designed and developed internal Python libraries and FastAPI-based microservices for ML model serving and automation.
- Designed and implemented an A/B testing analysis platform (Streamlit, BigQuery, statsmodels, custom bootstrap engine) enabling reliable experiment validation with 1-day result latency.
- Architected a cost-efficient GrowthBook query layer (Terraform, BigQuery, GA4) that achieved a 20x reduction in BigQuery processing costs for A/B test analytics.

Data Analyst - **Ecentria Inc. (OpticsPlanet.com)** *Feb 2018 - Jan 2021*

- Designed and maintained BI reports and dashboards (PowerBI, SQL, SSRS).
- Conducted quantitative analysis on large datasets to guide business strategy.
- Partnered with data engineers to optimize data pipelines and improve accessibility.

Financial Controller - **RISOIL S.A.** *2017 - 2018*

Financial Controller / Manager - **KONTI** *2004 - 2014*

EDUCATION

MSc, Engineering and Economics - **Odesa National University of Technology, Ukraine** *1999 - 2004*

LANGUAGES

English (Advanced) **Ukrainian** (Native)